MSAS – Assignment #2: Modeling Andrea D'Uva, 942644

1 Questions

Question 1

1) List the stages of dynamic investigation and their meaning. 2) When going from the *real system* to the *physical model* a number of assumptions are made; report the most important ones along with their mathematical implications. 3) For each of the assumptions below, shortly state what sort of simplification may result: i) The gravity torque on a pendulum is taken proportional to the pendulum angle θ; ii) Only wind forces and gravity are assumed in studying the motion of an aircraft; iii) A temperature sensor is assumed to report the temperature exactly; iv) The pressure in a hydraulic actuator is assumed uniform throughout the chamber. 4) List the *effort* and *flow* variables for the domains treated and discuss their similarity.

With dynamic investigation we refer to a specific approach used to model and simulate the behaviour of a real system we want to study or build. This method is composed of several distinguished steps, as listed below:

1. Specify the complex system under investigation, also denoted as *real system*.
2. From the real system we build the *physical model*, which is an abstraction of the reality and is developed to be as simple as possible while being as accurate as possible depending on the given requirements.
3. We proceed to formulate the *mathematical model* based on the physical one: the aim is to derive a set of mathematical equations on which we are now able to work.
4. Once we are in the mathematical world we represent the significant equations in a computational environment that allows us to solve them and compute the system response, either numerically or in an analytical form.
5. When the dynamic response has been obtained we can analyze it and make choices on some parameters or assumptions made at the beginning in order to get a better behaviour of the system.
6. As last step a kind of feed-back loop can be added. If experimental data are available from some testings over the real system we can compare them with the dynamic response we got to check for structural or parametric errors we have made. On top of it an outer loop can always be established such that at each iteration of the procedure we can refine more and more our design.

It is vital when developing the *physical model* to retain the most important characteristics of the real system without adding too much complexity which would only waste computational resources. Therefore, as engineers, is it crucial to make an abstraction using both our engineering experience and approximations.
Some of the most common approximations are:

- **Neglect small effects**: it helps simplifying the mathematics involved by discharging those phenomena that are negligible compared to main physics that rules the system behaviour.
- **Unaltered environment**: it is usually reasonable to assume that the environment surrounding the system is not affected by the latter. Again this helps in reducing the problem complexity because we can avoid coupling phenomena.

- **Lumped parameters approximation**: such an assumption must always be verified and can't be made mindlessly. That is because when we move from a distributed parameter system to a lumped parameter one we are switching the maths involved from PDEs to ODEs. It is not just a simplification but a substantial modification of the intrinsic nature of the system.

- **Linear cause-effect relationship**: once again the main goal is to simplify the *mathematical model* by not considering non-linear effects.

- **Time-invariant parameters**: we assume the physical parameters of the system to be constant over time. It might be useful when solving numerically our equations although strictly speaking it is not essential.

- **Neglect uncertainty and noise**: in the real world those effects are always present. However if we consider them from the beginning of the modelling phase we couldn't use anymore a deterministic approach: that's why usually these phenomena are taken into account only in the last stage of the design when a statistical analysis can be performed.

Considering some simple examples can help in understanding when and why approximations are introduced:

(i) The exact equation of motion for an ideal simple pendulum is $\ddot{\theta} + \frac{g}{L}\sin\theta = 0$ which implies facing an elliptic integral that doesn't have a closed form solution. Therefore to get a first physical insight it is usually assumed that the gravity torque is simply proportional to the pendulum angle θ. Such approximation actually means that we are linearizing the equation of motion around the equilibrium point $\theta = 0$, so the term $\sin\theta$ is approximated as θ and the equation of motion can be integrated analytically

(ii) In aircraft flight mechanics the only external forces considered relevant are gravity and aerodynamic loads. This doesn't represent the exact reality because there are many more physical phenomena affecting an aircraft behaviour, one of which might be the solar radiation pressure. However only the two mentioned forces are actually relevant, having all other ones an impact lower by several orders of magnitude. Thus this is a good example of small effects neglection.

(iii) In the real world all sensors, e.g. a temperature one, have to face with some noise and uncertainity coming from complex phenomena which behaviour cannot be predicted in advance. When we design a system that has some of these probes inside, we usually neglect those random effects so that we can approach the modelling phase through a deterministic approach instead of a statistical one that is much more cumbersome.

(iv) When approaching fluid systems like hydraulic actuators we first assume the pressure to be uniform in the control volume. This means that we lump a distributed parameter into a single value. Such approximation is useful for a first study of the behaviour, leaving the complex 3D fluid dynamics investigation to a successive step.

In our main engineering fields, namely the mechanical domain, the thermal domain, the electrical domain and the fluid domain, a similarity between the major quantities of interest can be found.
For each of them we can identify an *effort* variable and a *flow* variable: the first is related to the ability of doing work while the second refers to the rate of change at which work can be done.
Table 1 shows a possible choice we can make for each of the listed domains.

Such an abstraction in particularly useful because it allows us to identify three abstract components of our systems:

1. **resistor**: it's a component that applies the following algebraic $\textit{effort} = \textit{resistance} \times \textit{flow}$ and usually have a dissipative nature.
2. **capacitor**: it'a an element that can store energy by virtue of the *effort* variable and is characterized by the differential relation $flow = \textit{capacitance} \times \frac{\mathrm{d}\textit{effort}}{\mathrm{d}t}$.
3. **inductor**: can be interpreted as the opposite of a capacitor as energy is managed by the *flow* variable ad the differential relation is $\textit{effort} = \textit{inductance} \times \frac{\mathrm{d}\textit{flow}}{\mathrm{d}t}$.

Table 1: *effort* and *flow* variables for different engineering fields

Domain	*Effort*	*Flow*
Mechanical	Force	Velocity
Fluid	Pressure	Volumetric flow rate
Electrical	Voltage	Current
Thermal	Temperature	Heat flow

At this point it is also worth mentioning some exceptions to these general rules: the thermal domain doensn't have an inductance and it's resistance does not dissipate energy. Moreover also the mechanical domain shows a peculiar behaviour since differently from the other fields, the *flow* and *effort* variables do not have a unique choice, sometimes me may also need to swap them, so attention should be take when dealing with resistors, capacitors and inductors.

Question 2

1) Briefly discuss the physical meaning of the bulk modulus; show how the *effective* bulk modulus is computed. 2) Under what circumstances the fluid resistance yields a linear relation between effort and flow variables? 3) Find the expression of the leakage through a thin annular gap starting from the balance between shear stress and pressure drop.

Generally speaking liquids do not have a closed-form expression for their equation of state because their behaviour changes dramatically in wide ranges of p and T. What is usually done is linearizing the generic relation $\rho = \rho(p, T)$ around some reference conditions (p_0, T_0). The resulting expression is:

$$\rho(p,T) = \rho_0 \left[1 + \frac{1}{\beta}(p - p_0) - \alpha(T - T_0)\right]$$

Having defined the two coefficients as:

- **Bulk modulus**: $\beta = \rho_0 \left(\frac{\partial \rho}{\partial p}\right)_{p_0,T_0}$
- **Thermal expansion coefficient**: $\alpha = -\frac{1}{\rho_0}\left(\frac{\partial \rho}{\partial T}\right)_{p_0,T_0}$

In our typical aerospace hydraulic system the Bulk modulus is the coefficient that matters the most. If we express the density as $\rho_0 = \frac{m_0}{V_0}$ then its definition can be also written as $\frac{1}{\beta} = -\frac{1}{V_0}\left(\frac{\partial V_0}{\partial p}\right)_{p_0,T_0}$.
Therefore the *Bulk modulus* is a coefficient that refers to how the liquid volume changes when it is compressed at the reference initial conditions; it's a sort of liquid elasticy. By definition we have $\beta > 0$; it has the units of pressure, typically ranging between 1 and 2 GPa.
Moreover for fluids it also doesn't have an analytical expression, so to if we want an equation for $\beta = \beta\,(p_0, T_0)$ we need to rely on experiments and empirical laws.

In real-world hydraulic systems we don't have to take care of the liquid behaviour only: small concentrations of gas can occur and the container gets deformed when the pressure increases. All these effects are taken into account by means of the *effective bulk modulus* β_e. In order to define it we need to consider the volume and bulk modulus of each of those components.
By definition we can write the *effective bulk modulus* using finite variations instead of derivatives as: $\frac{1}{\beta_e} = \frac{\Delta V_t}{V_t \Delta p}$ being $V_t = V_l + V_g + V_c$ and $\Delta V_t = -\Delta V_l - \Delta V_g + \Delta V_c$, ith the $-$ sign in order to account for the negative volume variations of both fluid and gas.
Merging these equations and using the definitions to highlight $\frac{1}{\beta_l}$, $\frac{1}{\beta_g}$ and $\frac{1}{\beta_c}$, we get to the intermediate step

$$\frac{1}{\beta_e} = \frac{V_l}{V_t \beta_l} + \frac{V_g}{V_t \beta_g} + \frac{1}{\beta_c}$$

Then if we neglect V_c, we can express $V_l = V_t - V_g$ so we obtain the exact equation:

$$\frac{1}{\beta_e} = \frac{1}{\beta_l} + \frac{V_g}{V_t}\left(\frac{1}{\beta_g} - \frac{1}{\beta_l}\right) + \frac{1}{\beta_c}$$

However from experimental values we know that typically $\beta_l \gg \beta_g$, so we can further simplify the equation to get the operative one:

$$\frac{1}{\beta_e} \cong \frac{1}{\beta_l} + \frac{V_g}{V_t \beta_g} + \frac{1}{\beta_c}$$

The expression is now ready to be used, and will depend also on the formula we consider for the various components' *bulk modulus.*
For instance we always need an experimental law for β_l, while for a cylindric container we can usually apply the equation $\beta_c = \frac{Et}{D}$ and in case of perfect gas assumption we have by definition that $\beta_g = p$.

A major effect occurring in hydraulic networks is the so-called *pressure drop*. It is depicted clearly when applying the energy balance on a control volume under the following assumptions: stationary flow of an incompressible fluid, duct with constant section, no height difference and no energy introduced nor absorbed through the boundary. The energy conservation equation then takes the following form:

$$p_1 - p_2 = c_v(T_2 - T_1)$$

The fundamental message it sends is that even for a simply flowing liquid we might experience a transformation from pressure to thermal energy. This phenomena is labelled *Head loss*, and it always occur in real flows because of the friction action. Although being experienced as a temperature increase, we would like to express this phenomena by means of a connection between Δ*effort* and Δ*flow*, which might help in computing the amount of pressure loss.
To accomplish this goal is it very often used the Darcy-Weisbach law:

$$p_1 - p_2 = f\frac{1}{2}\frac{L}{D}\rho v^2$$

where L is the duct's length, $D = \frac{4area}{perimeter}$ is che hydraulic diameter, v is the average velocity and f is the friction factor. In general then we don't get the resistor relation $effort = resistance \times flow$ but instead a non linear one. The main complexity is hidden inside f, which depends on the nature of the flow, determined by the Reynolds number, and on the relative roughness ϵ.
Once again this coefficient in most situations has to be computed experimentally, only in a few cases we have an analytical expression available. One of this exceptions occurs when the fluid is flowing inside the duct in laminar regime, so characterized by $Re < 2000$ for a circular pipe. In this condition we can analytically find that $f = \frac{64}{Re}$, so globally:

$$\Delta p = \frac{32\mu L}{D^2}v$$

This case, along with the phenomena of laminar leakage through a thin annular gap, are the only two circumnstances in which the fluid resistance actually establishes a linear relation among Δ*effort* and Δ*flow*.

Let's now see how to analytically derive the expression for the resistance occurring during a flow inside a thin annular gap, as it usually happens in the case of a piston and a cylinder, shown in Figure 1a.
We have two paths we could follow: for both we need to start from basic fluid mechanics principles, however with one approach we need to work and integrate in cylindrical coordinates while with the other one we could derive a solution for a cartesian frame and after that just specialize the parameters involved.
The second way is simpler; therefore let's refer to the generic fluid cube depicted in Figure 1b, belonging to the leakage region. Let's assume stationary conditions and laminar flow: the latter means that the velocity has only the component directed along the flow and it varies only orthogonally to it, $v(y)$.
Given the symmetry of the problem we can focus only on the region $y > 0$, and the balance of momentum across two sections at a distance L reads:

$$A\,(p_2 - p_1) = -S\tau_w$$

where A is the cross-sectional area, S is the wet surface and τ_w is the friction at the wall. Therefore for the generic cube extending from 0 to y, and recalling that we can express $\tau = \mu \frac{dv}{dy}$, the explicit balance becomes:

$$yh\Delta p = -Lh\mu \frac{\mathrm{d}v}{\mathrm{d}y}$$

The velocity profile is then readily recovered by integrating from the generic y to the boundary at $\frac{e}{2}$ with the no-slip condition $v(\frac{e}{2}) = 0$:

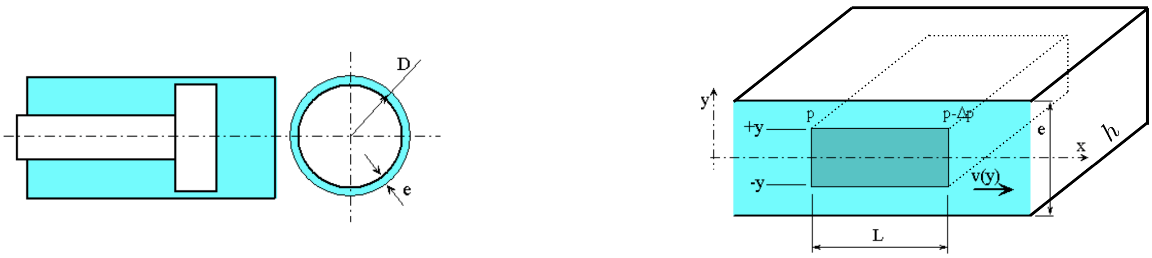
$$v(y) = \frac{\Delta p}{2L\mu}\left(\frac{e^2}{4} - y^2\right)$$

Now we can compute the total flow rate passing across the leakage section Q, being $dQ = v(y)hdy$, with a furthur integration:

$$Q = 2\int_0^{\frac{e}{2}} dQ = \frac{he^3\Delta p}{12L\mu}$$

Eventually, in case of a thin annular gap, the actual clearance is $g = 2e$ and we can approximate the width as $h = \pi D$ such that the final expression relating Q and Δp will be:

$$Q = \frac{\pi D g^3}{96L\mu}\Delta p$$



(a) leakage through a thin annular gap

(b) generic fluid cube inside the leakage section

Figure 1: Physical model of leakage through a thin gap

Question 3

1) Derive from scratch the mathematical model for RC and RL circuits and express the system response in closed form. 2) Consider a real DC motor and a) sketch its physical model (list the assumptions made); b) derive its mathematical model; c) show how the motor constant depends on the physical parameters.

Two of the simplest passive electrical circuits that have an internal dynamics are the RC and RL circuits.
Both are made by placing a voltage generator, the only active component in the network, a resistance and a capacitor or an inductor in series.
In both cases we are able to write a simple first-order mathematical model, which can be solved analytically given proper initial conditions.

Let's start analyzing the RC circuit depicted in Figure 2a; we need tthe following tools to develop the model: the Kirchhoff's voltage and current laws along with the characteristic equation for each component.
Assuming a constant generator, the voltage balance reads: $V_0 = V_R + V_C$ so across the resistor we have $V_R = V_0 - V_C$;
Then we can express the current flowing in the circuit using the constitutive laws of the resistor and the capacitor: $i = \frac{V_R}{R} = \frac{V_0 - V_C}{R}$ and $i = C\frac{\mathrm{d}V_C}{\mathrm{d}t}$.
Since the current is the same along the circuit we can set them equal:

$$\frac{V_0 - V_C}{R} = C\frac{\mathrm{d}V_C}{\mathrm{d}t}$$

that can be rewritten in a better way as

$$\frac{\mathrm{d}V_C}{\mathrm{d}t} + \frac{1}{RC}V_C = \frac{V_0}{RC} \tag{1}$$

Equation 1 is a first-order linear ordinary differential equation with constant coefficients in the unknown $V_C(t)$ that can be solved in a closed-form if $V_C(0)$ is provided.
Assuming, without loss of generality, that the capacitor is initially unloaded, i.e. $V_C(0) = 0$ V, and that the switch closes at $t_0 = 0$; we first derive the homogenous solution by neglecting V_0 and assuming $V_C^{\mathrm{hom}} = Ae^{\lambda t}$.
The equation yields $\lambda = -\frac{1}{RC}$ such that $V_C^{\mathrm{hom}} = Ae^{-\frac{t}{RC}}$.
Regarding the particular solution $V_C^p(t)$, we can immediately notice that the forcing term is a constant, thus it will simply be $V_C^p(t) = V_0$.
Finally we can sum them and then use the initial condition to determine the value of the integration constant A so the system response will be:

$$V_C(t) = V_0\left(1 - e^{-\frac{t}{RC}}\right) \tag{2}$$

From Equation 2 we can immediately notice that the voltage across the capacitor will tend to the value V_0, when it will be fully charged and the current will stop flowing.
Defining the *time constant* as $\tau = RC$, we usually say that the transient will be over after a time of about $4 \div 5\tau$ s.
To find also the current evolution across the circuit we can simply use the characteristic equation of the resistor.

The analysis of the RL circuit shown in Figure 2b applies the very same concepts.
The characteristic equation of the inductor is $V_L = L\frac{\mathrm{d}i}{\mathrm{d}t}$, while the resistor's one is again $V_R = Ri$.

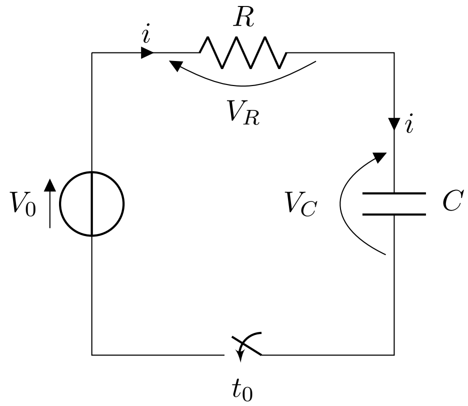
Now we apply the Kirchhoff's voltage law on the network to get $V_0 = V_R + V_L$.
Merging these three equations allows us to compute the ode describing the circuit response:

$$\frac{\mathrm{d}i}{\mathrm{d}t} + \frac{R}{L}i = \frac{V_0}{L} \tag{3}$$

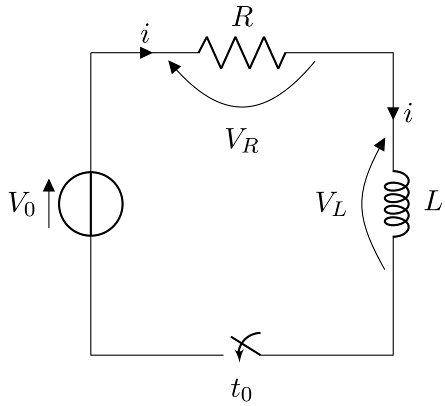
Equation 3 rules the current evolution in time across the system; its nature is the very same as Equation 1.
In order to get a closed form solution let's assume again that the circuit is initially unloaded, i.e. $i(0) = 0$ A, and that the switch closes at time $t_0 = 0$.
Proceding exactly as before we can first find the homogenous solution $i^{\text{hom}} = Be^{-\frac{Rt}{L}}$ and the particular one $i^p = \frac{V_0}{R}$.
Taking their sum and using the initial condition delivers the system response in terms of the current:

$$i(t) = \frac{V_0}{R}\left(1 - e^{-\frac{Rt}{L}}\right) \tag{4}$$

As Equation 4 shows current flow then will grow in time until it reaches the steady-state where $i(t) = \frac{V_0}{R}$, because the inductor behaves like a short circuit when the transient has expired.
Again we can define a *time constant* $\tau = \frac{L}{R}$, and we usually say that the system reaches the equilibrium after a period of about $4 \div 5\tau$ s.



(a) RC circuit



(b) RL circuit

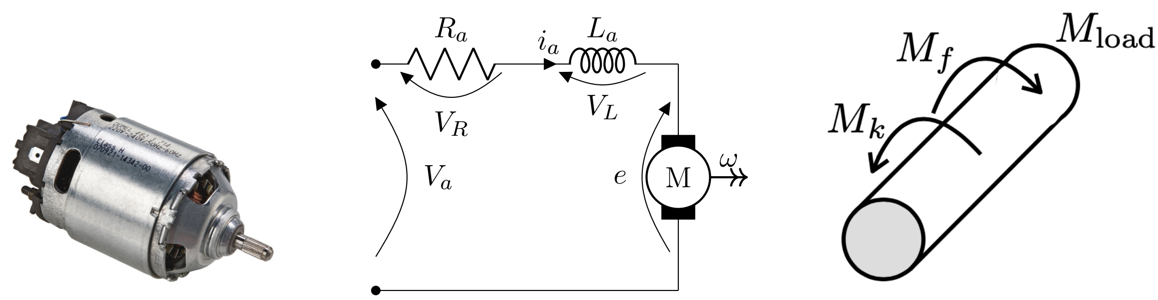
Figure 2: Physical models of two trivial electrical circuits exhibiting a transient behaviour

Electromechanical machines are devices that can convert electric energy into mechanical energy and viceversa. In the first case they are called *electric motors* and in the other scenario they are referred to as *electric generators.*
In both cases the generic construction is the same: we find a stator, which is a still part devoted to the generation of a magnetic field, and a rotor, or armature, which is the moving part connected to the shaft representing the mechanical side of the machine. Also equations are almost the same, the main difference being the signal considered as input: it will be the provided electric parameters in case of motors and the given mechanical load when dealing with a generator.
In these machines we usually find two different electrical circuits:

- **Excitation circuit**: placed on the stator and devoted to the magnetic field generator. Can be replaced by permanent magnets in some devices.
- **Armature circuit**: it's the circuit on the rotor and it is the one that feels the magnetic flux.

Depending on how the current is provided to the circuits we can classify three kinds of machines: DC machines, AC synchronous machines and AC induction machines.
Let's focus on a DC motor to try and model it. The peculiarity of this class of devices is that all circuits in the system are feeded by DC current.
Even among DC motors there is a wide variety of machines; for simplicity we will study a DC electric motor with permanent magnets, as the one shown in Figure 3a.
In this class of devices the excitation circuit is not present and the magnetic field is created by magnets placed in the stator. This feature leads to smaller motors which are also easier to be modelled and controlled; the disadvantage is that the magnetic field cannot be regulated.

In order to derive a physical model of this machine we can focus just on the electric circuit of the rotor and on the mechanical representation of the shaft.
To simplify the following assumptions are made: the air gap does not influence the behaviour of the magnetic flux density B, hysteresis and eddy currents in the material of the rotor are considered negligible, the only source of mechanical loss comes from the bearings and the brushes, the shaft is considered to be a rigid body, we will use a lumped-parameters approach and we have an ideal conversion between electric and mechanical energies.
The physical model of the armature circuit is depicted in Figure 3b: the control variable is the voltage V_a, R_a is the lumped resistance of the coil and the inductor L_a accounts for the self-inductance of the circuit. Moreover we have to consider the opposing electromotive force e that arises from the Faraday-Lenz phenomena.
The shaft is modelled as in Figure 3c: it is a rigid body subjected to the torque M_k generated by the motor, to the external load M_{load} and to the torque M_f which sums all the frictional effects, modelled as proportional to the velocity. It is characterized by the angular velocity ω and by the rotational inertia J.



(a) Real DC motor with permanent magnets

(b) Physical model of the armature circuit

(c) Physical model of the motor shaft

Figure 3: Real and physical models of a DC motor

Obtaining the mathematical model is then straightforward: we just need to apply the Kirchhoff's voltage law on the armature circuit and the momentum balance on the rigid shaft, recalling the characteristic equations of an electrical motor $e = K\omega$ and $M_k = Ki$ being K the motor constant. Thus the set of coulped ordinary differential equations ruling the motor dynamics is:

$$\begin{cases} V_a = R_a i + L_a \dfrac{\mathrm{d}i}{\mathrm{d}t} + K\omega \\ J\dfrac{\mathrm{d}\omega}{\mathrm{d}t} = Ki - b\omega + M_{load} \end{cases} \tag{5}$$

The last step of the modelling phase involes a deeper understanding of the motor constant K. In order to do that we should investingate the electric and magnetic interactions occurring between the stator and the rotor.
Let's consider the single turn immerged in a constant magnetic field depicted in Figure 4: the condition is similar to what happens on a permanent magnet DC motor where $\mathbf{B}$ is assumed constant.
The Lorentz law states that a conductor with a current flow inside a magnetic field will experience a force $\mathbf{F} = i\mathbf{l} \wedge \mathbf{B}$.
Therefore in our generic rectangular turn there are only 2 sides that produce a net torque and they are those sketched in Figure 4b. Along each of those segment the net force generated is equal to $F = ilB$ but the only component generating a moment is $F_r = ilB \sin\alpha$, where we might also express $\alpha = \omega t$ for a better temporal study of the generated force.
The torque produced by a single coil made of N turns, at the reference angle $\alpha = 90°$, can be computed as

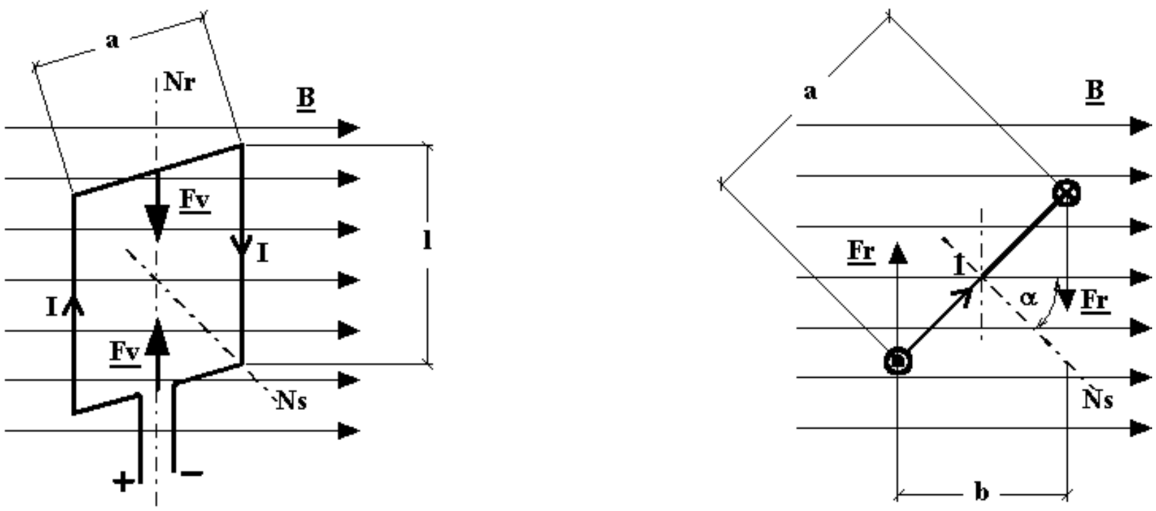
$$T_m = 2 \cdot NF_r \cdot \frac{a \sin\alpha}{2} = NalBi = K_T i \tag{6}$$

The magnitude of the counterelectromotive force e can be expressed in terms of the magnetic flux ϕ by means of the Farady law: $e = \frac{\mathrm{d}\phi}{\mathrm{d}t}$. In our specific motor both the magnetic field density B and the coil shape are constant, so the magnetic flux will depend only on the inclination coil-field, i.e. $\phi = \phi(\alpha)$.
Thus we can state

$$e = \frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{\partial\phi}{\partial\alpha}\frac{\mathrm{d}\alpha}{\mathrm{d}t} = K_e\omega \tag{7}$$

Now, the mechanical power is $P_M = T\omega$ while the electric power is given by: $P_E = ei$. Assuming an ideal conversion we have $P_M = P_E$ which becomes $K_T i\omega = K_e i\omega$.
Eventually we found that in our case we have a single motor constant which depends the physical parameters N, a, l, B:

$$K_T = K_E = K = NalB \tag{8}$$



(a) View from above

(b) View aligned with the coil

Figure 4: Different views of a turn immerged in a magnetic field

Question 4

1) Write down the Fourier law and show how it is specialized in the case of conduction through a thin plate; discuss the concept of thermal resistance. 2) Report the equation for thermal radiation in case of a) black body and b) real body and discuss them.

Heat can be transferred in three ways: conduction, convection and radiation.
When we speak of conduction we refer to the ability of a still continuos media, usually a solid, to propagate thermal energy. This physical mechanisms is modelled through a fundamental principle, named Fourier law, which states:

$$\frac{Q_h}{A} = -k_t \nabla T \tag{9}$$

This is the general three dimensional version of the law which relates the *heat flow* $Q_h\,[W]$ to the temperature gradient $\nabla T\,\left[\frac{K}{m}\right]$. In general both these quantities are functions of space and time.
The other quantities involved in the law are the cross-sectional area $A\,\left[m^2\right]$ and the thermal conductivity $k_t\,\left[\frac{W}{mK}\right]$, which is an intrinsic property of the material and in general is itself a function of the temperature.
Note also the presence of the $-$ sign: heat propagates along the direction of the temperature drop.
It is then clear that Equation 9 is a non-linear partial differential equation which makes the dynamic analysis of thermal systems much more complex: they inherently are distributed parameters systems and the solution might also be impossible to get if the geometry is not simple enough.
However e can specialize the Fourier law for some peculiar conditions and geometries and obtain basic but still useful solutions.
Let's refer to the case of conduction through a flat plate of length L and cross-sectional area A as depicted in Figure 5. If we can assume that the temperature drop occurs mainly along one

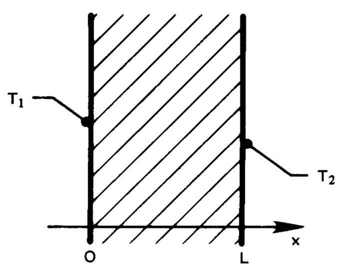


Figure 5: Conduction through a thin flat plate

direction, the Fourier law takes its one dimensional version:

$$\frac{Q_h}{A} = -k_t \frac{\mathrm{d}T}{\mathrm{d}x} \tag{10}$$

To further simply the tractation by assuming also that the thermal conductivity of the material k_t is constant and that the heat flow Q_h is uniform across the layer. The latter assumption actually means that we are considering a steady-state problem and that no heat sources are present inside the flat plate.
We can now proceed and integrate Equation 10:

$$\int_0^L \frac{Q_h}{A}\,dx = -\int_{T_1}^{T_2} k_t(T)\,dT \tag{11}$$

and we obtain the resulting equation

$$Q_h = \frac{k_t A}{L} \Delta T \tag{12}$$

being $\Delta T = (T_1 - T_2) > 0$.
Recalling that for thermal domain we defined the effort variable as the temperature T and the flow variable as the heat flow Q_h, if we rearrange Equation 12 as $\Delta T = \frac{L}{k_t A} Q_h$, we can notice the similarity with the algebraic expression *effort* = *resistance* × *flow*.
This allows us to define the concept of thermal resistance

$$R = \frac{L}{k_t A}$$

The concept of thermal resistance is extremely useful for several reasons: since we assumed Q_h to be uniform then across a flat plate the temperature distibution will be linear, moreover it behaves similarly to electric resistances and hence when modelling a system we can consider series or also parallel resistances.
This property of being able to solve thermal circuits similarly to electrical ones is very often named *electric analogy*.
However we must always be careful we dealing with the concept of thermal resistances.
First of all we shall remember that contrary to the electric case, these elements do not dissipate any power; that is because the heat flow is already a power itself.
More importantly we need to recall the underlying assumptions that lead to the definition of thermal resistance: if any of these isn't valid then the analogy cannot be applied anymore.
For instance if inside a flat wall we have a source of thermal energy then, even if we are in steady-state conditions, modelling the system through a resistance would be a big mistake: the actual temperature distribution will be remarkably different.

Thermal radiation is another mechanism of heat propagation. Its main peculiarity is that physically the energy exchange occurs by means of electromagnetic waves, therefore unlike conduction and convection there is no need to have a medium between bodies as it can also take place in vacuum.
This topic needs complex physical and mathematical concepts to be treated in details, so only a short and simplified discussion will be held.
Let's first introduced the concept of *black body*: a black body, depicted in Figure 6 is an ideal object that is able to absorb all the incoming radiation. A black body is then considered the idealization of both the perfect absorber and emitter.

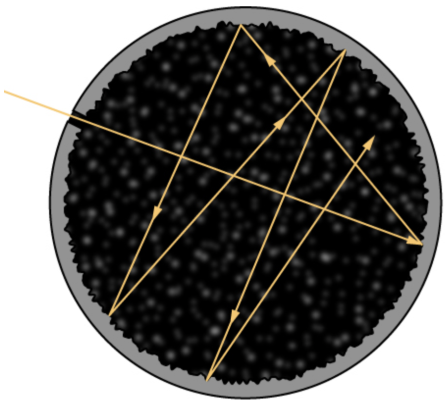


Figure 6: Representation of a *black body*: all incoming radiation is absorbed

Experimental results and mathematical analysis show that a black body radiates a heat flow that obeys the following law:

$$\frac{Q_h}{A} = \sigma T^4 \tag{13}$$

Where A is the external surface area of the object and σ is the Stefan-Boltzmann constant, $\sigma = 5.67 \cdot 10^{-8} \left[\frac{W}{m^2 K}\right]$.
However real bodies cannot emit all that power, thus we introduce a correction term called *emissivity* and denoted as F_e or ϵ. The emissivity takes the value 1 when referring to a black body, and value between 0 and 1 for real corpses.
Another important parameter we need to consider when dealing with radiation is the *view factor* F_v which expresses how much of the emitted enrgy actually impinges another body.
Computation of the view factor is usually a cumbersome task which involes either analytical calculations or numerical techinques.
Having introduced these fundamental parameters, the net radiative heat transfer between two bodies is computed as follows:

$$Q_h = F_e F_v \sigma A \left(T_H^4 - T_L^4\right) = \frac{\sigma \left(T_H^4 - T_L^4\right)}{\frac{1-\epsilon_1}{\epsilon_1 A_1} + \frac{1}{A_1 F_{1,2}} + \frac{1-\epsilon_2}{\epsilon_2 A_2}} \tag{14}$$

Equation 14 is the general rule that considers two real bodies, having labelled as H the higher temperature and as L the lower one. The case in which one or both corpses can be assumed to be black bodies is simply retrieved by setting $\epsilon_{1/2} = 1$.
The radiation equation is quite interesting as it is noticeably different from equation ruling the other mechanisms of heat transfer.
It's main feature is that it doesn't depend on ΔT but on the difference between the fourth power of each temperature.
This makes the equation instrinsically non-linear and thus we can't define a proper thermal resistance in analogy to the conduction and convection phenomena.
Moreover the presence of the constant σ, which has a small magnitude, means that radiation isn't always a dominant effect. For instance when we deal with bodies that have a similar temperature, and in presence of other energy exchanges, radiation is usually neglected.
If on the contrary the difference in temperature grows significantly, as it happens with spacecrafts outside the atmosphere, such a phenomena becomes more and more important and might also be the main ways through which heat is transferred.

2 Exercises

Exercise 1

A miniaturized reaction wheel can be modeled as a couple of massive disks, connected with a flexible shaft (Figure 7). The first disk is driven by a rigid shaft, linked to an electric motor. The motor provides a variable torque, while the rigid shaft is subjected to viscous friction, due to motor internal mechanisms. At $t_0 = 0$, the motor provides the torque $T(t) = T_0$.
1) Write down the mathematical model from first principles. 2) Using the data given in the figure caption, and guessing a value for the flexible shaft stiffness k and the viscous friction coefficient b, compute the system response from t_0 to $t_f = 10$ s. 3) Two accelerometers are placed on the two disks recorded samples at 100 Hz, which were saved in the file `samples.txt`; the samples are affected by measurement noise. Determine the value of k and b that allows retracing the experimental data, so avoiding parametric errors.

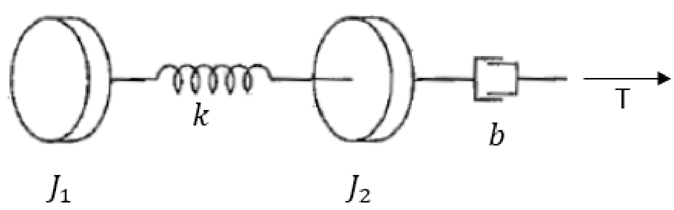


Figure 7: Physical model ($J_1 = 0.2\,\text{kg}\cdot\text{m}$; $J_2 = 0.1\,\text{kg}\cdot\text{m}$; $T_0 = 0.1\,\text{Nm}$).

The mathematical model for the physical system under investigation can be retrieved by separately applying the momentum equation to each rigid disk.
Disk 1 is subjected only to an elastic torque given by $M_e = k\,(\theta_2 - \theta_1)$, assuming without loss in generality that $\theta_2 > \theta_1$.
On disk 2, in addition to the same M_e, we also have the external torque T_0 and the viscous fricton, assumed at first to be linearly proportional to the angular velocity, $M_f = b\dot{\theta}_2$.
Thus, the governing equations can be written as:

$$\begin{cases} J_1\ddot{\theta}_1 = k\,(\theta_2 - \theta_1) \\ J_2\ddot{\theta}_2 = -k\,(\theta_2 - \theta_1) + T_0 - b\dot{\theta}_2 \end{cases} \tag{15}$$

The system 15 contains two second order ordinary differential equations with constant coefficients. Therefore four states can be identified, and the chosen ones have been $\left[\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2\right]^T$.
To get a first simulation of the system response, the damping coefficient was arbitrarily assumed to be $b = 4\left[\frac{Nrad}{s^2}\right]$ while the circular shaft was chosen to be made in a generic mild steel, $G = 75\,[GPa]$, with a length of $4\,[m]$ and a diameter of $25\,[mm]$.
Extimating the spring stiffness as $k = \frac{GI_{xx}}{L} = 44.95\,[Nrad]$, Figure 8 shows the transient response of the system.
It is physically clear that the states are unstable: while the angular velocities will eventually reach an asymptotic value, the angles θ_1 and θ_2 will increase indefinetely, unless a mathematical bound is established.
In order to preserve this unstable behaviour an explicit Runge-Kutta scheme has been adopted to numerically solve the equations of motion.

Then, given a series of experimental samples for the two accelerations, the strategy adopted to select a couple (k, b) was to first define a proper measure of error. Knowing that these samples were affected by measurement noise, the relative error measure has been chosen to be

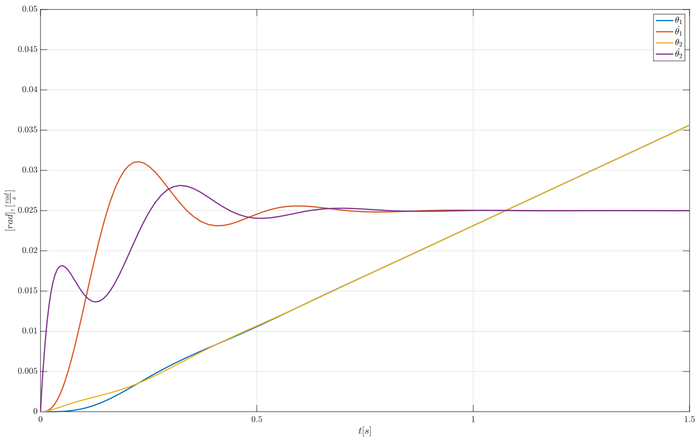


Figure 8: Transient response of the mechanical system

2-norm of a properly defined vector: $\|numericalresults - experimentalresults\|_2$.
The reason for this choice is that the 2-norm provides a measure that is similar to the standard deviation, so measured along each coordinate of the vector.
Figure 9 shows the point of minimum error and its coordinates, while Figure 10 shows the comparison between numerical solution and experimental samples with k, b selected from the search grid.

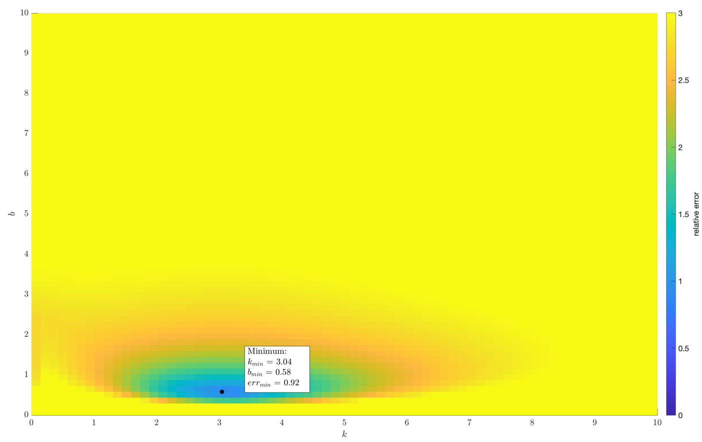
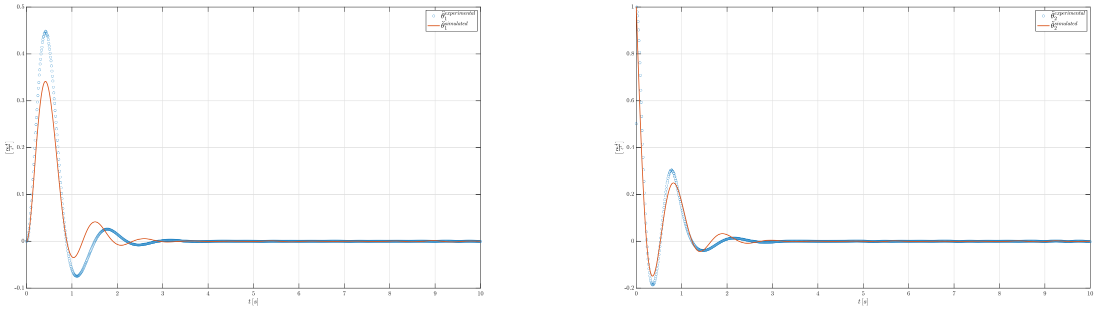


Figure 9: Relative error for couples of k, b

Being these results not satisfactory at all, the next iteration of the model saw the adoption of a non-linear damping torque:

$$\begin{cases} J_1\ddot{\theta}_1 = k\left(\theta_2 - \theta_1\right) \\ J_2\ddot{\theta}_2 = -k\left(\theta_2 - \theta_1\right) + T_0 - b\dot{\theta}_2\|\dot{\theta}_2\| \end{cases} \tag{16}$$

Then the search was conducted again according to the numerical results coming from the set of equations 18.
As Figures 11 12, this change in the model proved to be effective: data provided by the accelerometers are now well matched.



(a) Comparison for $\ddot{\theta}_1$ **(b)** Comparison for $\ddot{\theta}_2$

Figure 10: Numerical and experimental results

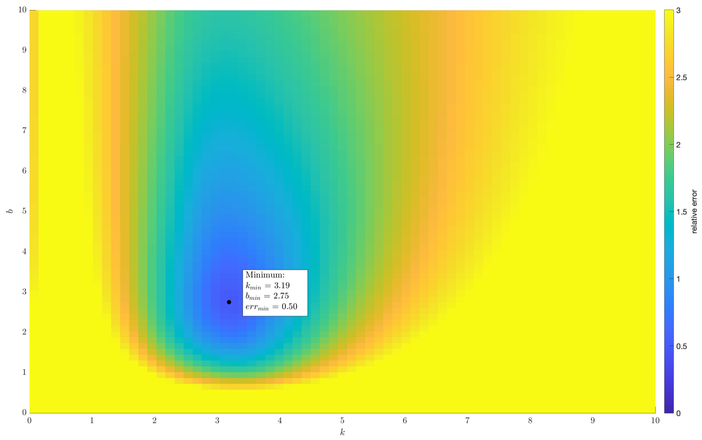
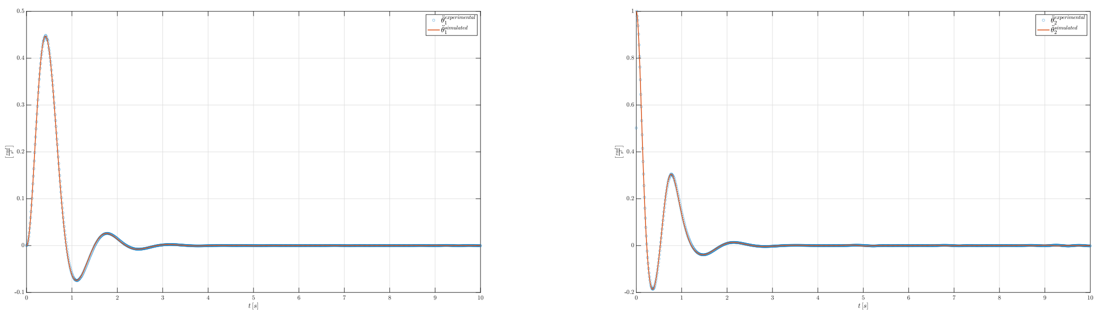


Figure 11: Relative error for couples of k, b



(a) Comparison for $\ddot{\theta}_1$ **(b)** Comparison for $\ddot{\theta}_2$

Figure 12: Numerical and experimental results for the second iteration

Exercise 2

The hydraulic system in Figure 13 consists of an accumulator, a check valve, a distributor, an actuator, and a tank, plus delivery and return lines. At $t = t_{-\infty}$, the accumulator contains nitrogen only. To charge it, the nitrogen undergoes an isothermal transformation from $\{p_{N_2}(t_{-\infty}), V_{N_2}(t_{-\infty})\}$ to $\{p_{N_2}(t_0) = p_0, V_{N_2}(t_0) = V_0\}$, t_0 being the initial time. 1) Assuming incompressible fluid, adiabatic discharge of the accumulator, and no leakage in the actuator, write down a mathematical model that allows computing pressures and flow rates in the sections labeled. 2) Considering the distributor command z in Figure 13, carry out a simulation to show the system response in $[t_0, t_f]$. 3) Determine the time t_e that takes the piston to reach the maximum stroke, $x(t_e) = x_{\max}$, starting from $x(t_0) = x_0$, $\dot{x}(t_0) = v_0$.

- Fluid: Skydrol, $\rho = 890$ kg/m^3.
- Accumulator: $V_{N_2}(t_{-\infty}) = 10$ dm^3, $p_{N_2}(t_{-\infty}) = 2.5$ MPa, $p_0 = 21$ MPa, adiabatic exponent $\gamma = 1.2$.
- Delivery: Coefficient of pressure drop[1] at accumulator outlet $k_A = 1.12$, coefficient of pressure drop across the check valve $k_{cv} = 2$, diameter of the delivery line $D_{23} = 18$ mm; Branch 2–3: Length $L_{23} = 2$ m, friction factor[2] $f_{23} = 0.032$.
- Distributor: Coefficient of pressure drop across the distributor $k_d = 12$, circular cross section, diameter $d_o = 5$ mm.
- Actuator: Diameter of the cylinder $D_c = 50$ mm, diameter of the rod $D_r = 22$ mm, mass of the piston $m = 2$ kg, maximum stroke $x_{\max} = 200$ mm; Load: $F(x) = F_0 + kx$, $F_0 = 1$ kN, $k = 120$ kN/m.
- Return: Diameter of the return line $D_r = 18$ mm; Branch 6–7: Length $L_{67} = 15$ m, friction factor $f_{67} = 0.035$.
- Tank: Pressure $p_T = 0.1$ MPa, initial volume $V_T(t_0) = 1$ dm^3, coefficient of pressure drop at tank inlet $k_T = 1.12$.
- Initial time: $t_0 = 0$, $x_0 = 0$, $v_0 = 0$; $t_1 = 1$ s, $t_2 = 1.5$ s; final time $t_f = 3$ s.

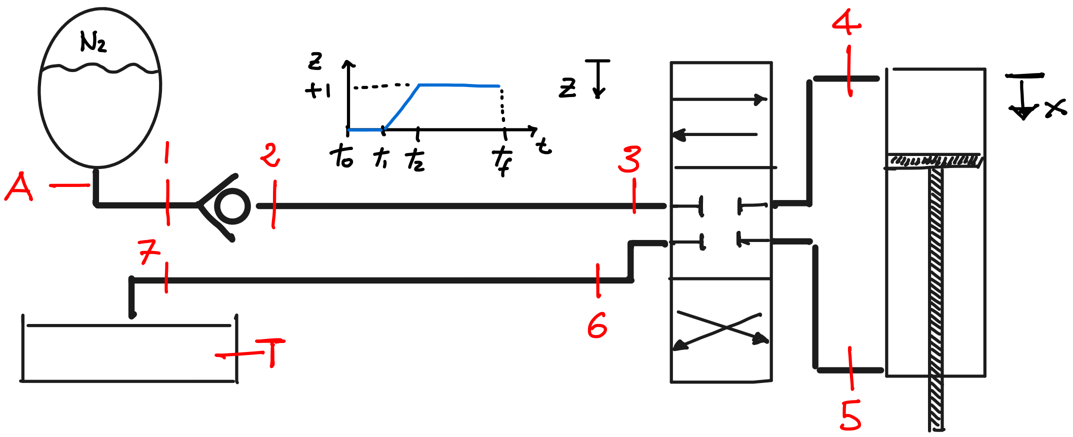


Figure 13: Hydraulic system physical model; assume any other missing data.

[1] $\Delta p = 1/2 k \rho v^2$.
[2] $\Delta p = fL/D 1/2 \rho v^2$.

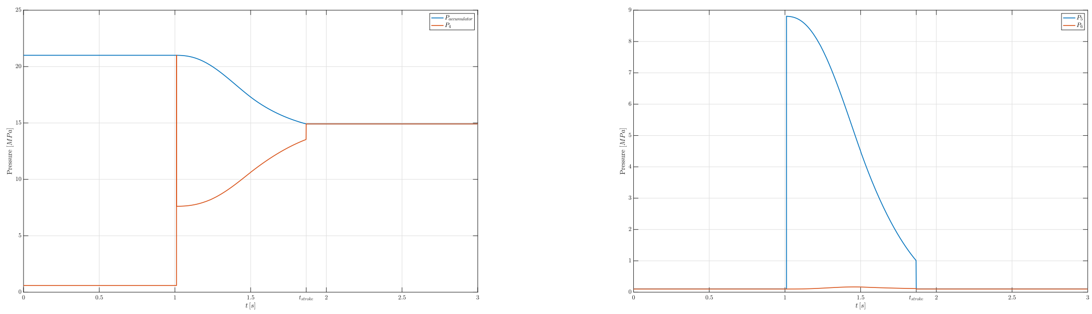
The hydraulic system, given all the assumptions made, has four evolving states; two are the accumulator and tank volumes and the are two are related to the actuator mechanics.
Their behaviour can be modelled as follows:

$$\begin{cases} \dot{V_{acc}} = Q_{accu} \\ \dot{x_{act}} = v_{act} \\ \dot{v_{act}} m_{piston} = P_4 A_4 - P_5 A_5 - F(x) \\ V_t \dot{a} nk = Q_{tank} \end{cases} \tag{17}$$

In addition we have also a number of algebraic equations that can be specialized considering the proper sections:

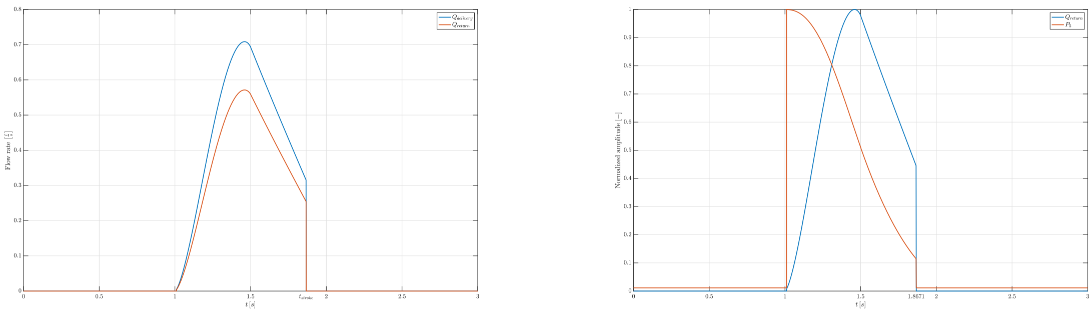
$$\begin{cases} \Delta P = \frac{1}{2} k \rho \frac{Q^2}{A^2} \\ \Delta P = \frac{1}{2} f \rho \frac{L}{D_h} \frac{Q^2}{A^2} Q_5 = A_5 v_{piston} \\ Q_4 = A_4 v_{piston} \\ P_{acc} = \left(\frac{P_0}{V_0 + V_{acc}} \right)^{\gamma} \end{cases} \tag{18}$$

It should also be noticed that the pressure of the accumulator is what actually rules the system, flow rates are constant along the delivery and return lines and the tank pressure is assumed to be constant.
Figures 14, 15a, 16 show the behaviour in time for the main quantities of interest in the network.
It is important to notice those sudden horizontal lines representing discontinuities.
There are two main explanations for such a behaviour: first of all fluid-mechanical system are inherently stiff being the fluid inertia much lower than the mechanical one.
Moreover, having assumed an incompressible liquid, we don't have pressure waves anymore but the speed of sound is infinite and thus each variation is immediately felt along the whole network.
That's why for the two main occuring events, the opening of the distributor and the piston stroke, such discontinuities cannot be avoided.



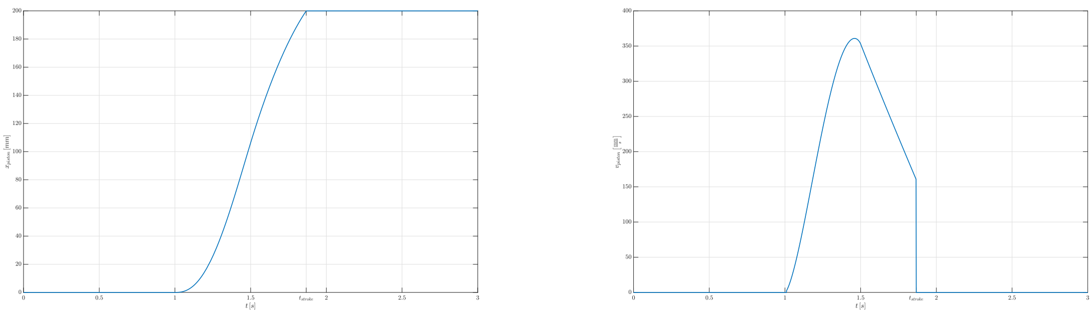
(a) **(b)**

Figure 14



(a) (b)

Figure 15



(a) (b)

Figure 16

Exercise 3

Consider the ideal physical model network shown in Figure 17. The switch has been open for a long time. The capacitor is charged and has a voltage drop between its ends equal to 1 V. Then, at $t = 0$, the switch is closed.

1) Plot the subsequent time history of the voltage V_C across the capacitor.
2) Assume a voltage source characterized by $v(t) = \sin(2\pi f t)\arctan(t)$ having the positive terminal downward inserted in place of the switch. What is in this case the voltage history across the capacitor?

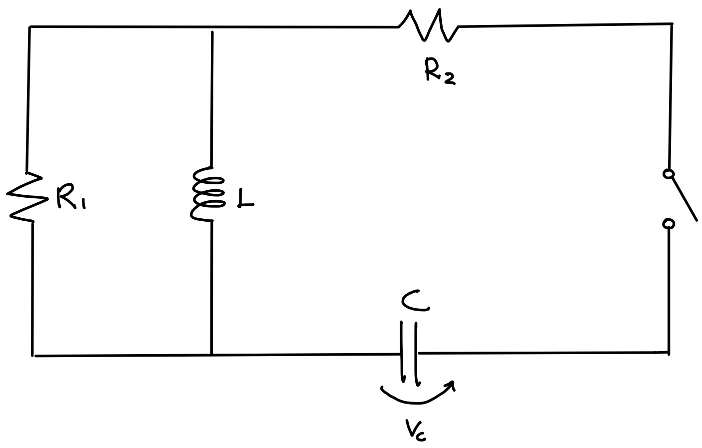


Figure 17: Circuit physical model ($R_1 = 1000\,\Omega$; $R_2 = 100\,\Omega$; $L = 1\,\text{mH}$; $C = 1\,\text{mF}$; $f = 5\,\text{Hz}$.)

To get the matematical model we need to exploit the Kirchhoff's voltage and current laws, taht for this circuit are expressed as:

$$\begin{cases} i_c = i_{R_1} + i_L \\ V_c + V_{R2} + V_L = 0 \\ V_C + V_{R2} + V_{R1} = 0 \end{cases} \tag{19}$$

Moreover we have to consider the consitutive equations for each component:

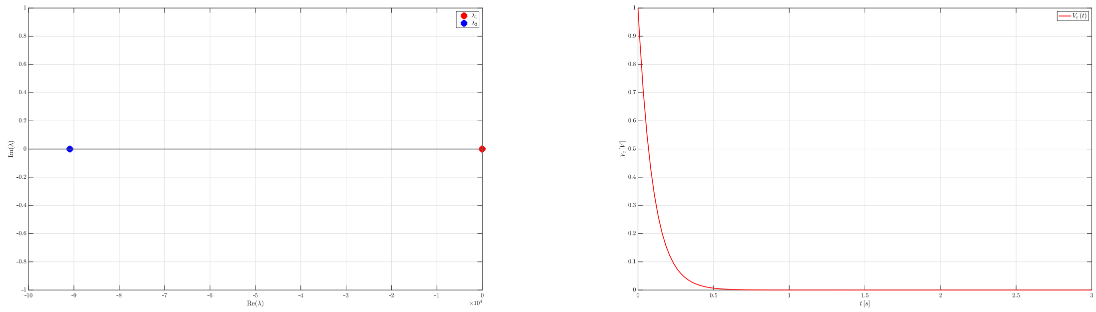
$$\begin{cases} V_{R1} = i_1 R_1 \\ V_{R2} = i_2 R_2 \\ i_C = C\dfrac{\mathrm{d}V_c}{\mathrm{d}t} \\ V_L = L\dfrac{\mathrm{d}i_l}{\mathrm{d}t} \end{cases} \tag{20}$$

From these seven equations we are able with just some simple substitutions, to get the equation:

$$\ddot{V}_c LC\left(1 + \frac{R_2}{R_1}\right) + \dot{V}_C\left(R_2 C + \frac{L}{R_1}\right) + V_C = 0 \tag{21}$$

As Figure 18a show, the equation has two real negative poles: the system is stable but is higly stiff as they differ by several orders of magnitude.
There an implicit A-stable method is required to properly integrate it.

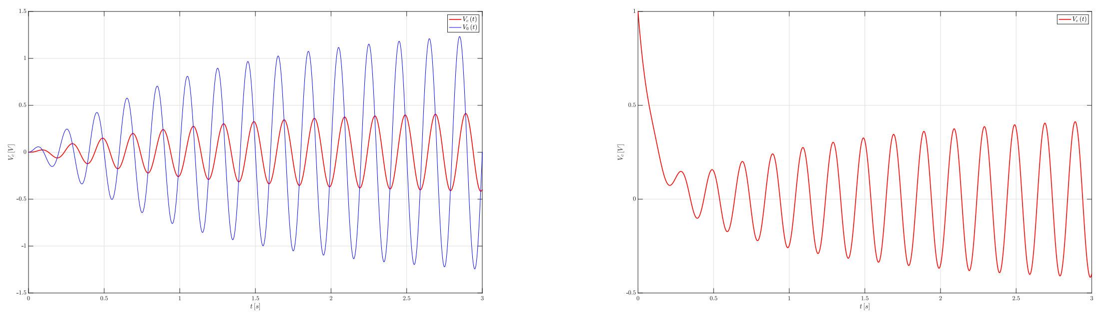
When a voltage source is inserted, the previous equation is only modified by the presence of the term $\frac{V_0}{LC} - \frac{\dot{V}_0}{R_1 C}$ which acts as forcing term.
However the numerical integration still works fine and FIgure 19 shows the oscillating behaviour induced by the time-variable voltage source.



(a) Eigenvalues of the system

(b) Free response

Figure 18



(a) The response has a phase shift

(b) Forced response with non-homogenous solutions

Figure 19: Forced response of the system

Exercise 4

The rocket engine in Figure 20 is fired in laboratory conditions. With reference to Figure 20, the nozzle is made up of an inner lining (k_1), an inner layer having specific heat c_2 and high conductivity k_2, an insulating layer having specific heat c_4 and low conductivity k_4, and an outer coating (k_5). The interface between the conductor and the insulator layers has thermal conductivity k_3. 1) Select the materials of which the nozzle is made of[3], and therefore determine the values of k_i ($i = 1, \ldots, 5$), c_2, and c_4. Assign also the values of ℓ_i (i=1,...,5), L, and A in Figure 20. 2) Derive a physical model and the associated mathematical model using one node per each of the five layers and considering that only the conductor and insulator layers have thermal capacitance. The inner wall temperature, T_i, as well as the outer wall temperature, T_o, are assigned. 3) Using the mathematical model at point 2), carry out a dynamic simulation to show the temperature profiles across the different sections. At initial time, $T_i(t_0) = T_o(t) = 20$ C°. When the rocket is fired, $T_i(t) = 1000$ C°, $t \in [t_1, t_f]$, following a ramp profile in $[t_0, t_1]$. Integrate the system using $t_1 = 1$ s and $t_f = 60$ s. 4) Repeat the simulation in point 3) using a mathematical model implementing two nodes for the conductor and insulator layers.

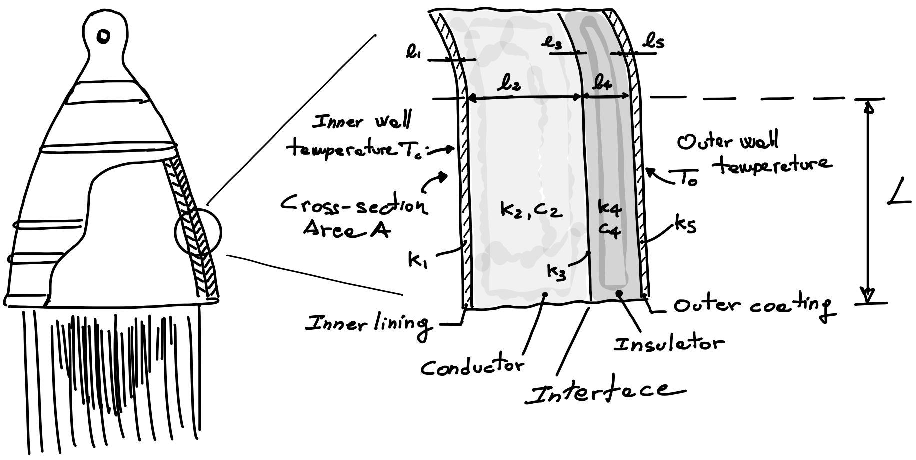


Figure 20: Real thermal system.

The materials chosen for the five different layers are actually three: graphite, silica aerogel and a generic type of ultra-high temperature ceramics. They have been chosen mainly by condidering their heat transfer properties, without taking into account any kind of mechanical behaviour, as it would be the case in a real-world design scenario.
Graphite has been selected for it's relatively high thermal conductivity. Its assumed properties are: $\rho = 1500 \left[\frac{kg}{m^3}\right]$, $k_t = 100 \left[\frac{W}{mK}\right]$ and $c = 750 \left[\frac{J}{kgK}\right]$.
Silica aerogel is taken to be the insulator medium, being characterized by $\rho = 150 \left[\frac{kg}{m^3}\right]$, $k_t = 0.015 \left[\frac{W}{mK}\right]$ and $c = 1000 \left[\frac{J}{kgK}\right]$.
Ceramics was employed in the tree smaller layers providing thermal contact and has a thermal conductivity of $k_t = 30 \left[\frac{W}{mK}\right]$.
The main advantage given by this choice is that their melting point is above $1000°C$.
Focusing on the lengths, their have been arbitrarily selected as $7, 50, 7, 50, 7\,[mm]$; the values of A and L actually don't influence the results of the model, so they have been considered as

[3] The interface layer is not made of a physically existing material, though it produces a thermal resistance. For this layer, the value of the thermal resistance R_3 can be direcly assumed, so avoiding to choose k_3 and ℓ_3.

unitary quantities just to make sure that a 1D conduction is actually a good description of the process.

In order to get a physical model, a completely one dimensional heat transfer has been assumed; moreover the thermal conductivities were kept constant, so indipendent from the local temperature.
The two resulting models, that consider respectively one and two lumped elements per layer, are depicted in Figure 21 and Figure 22, and they exploit the electrical analogy for thermal systems.

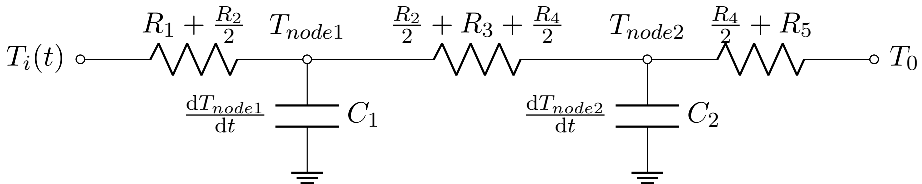


Figure 21: Physical model of a 2 lumped nodes thermal system using the electrical analogy

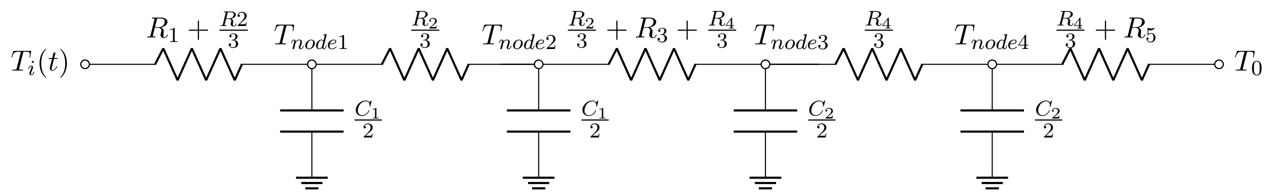


Figure 22: Physical model of a 4 lumped nodes thermal system using the electrical analogy

To retrieve the mathematical model, the main tool is to apply to each capacitive node the balance of powers:

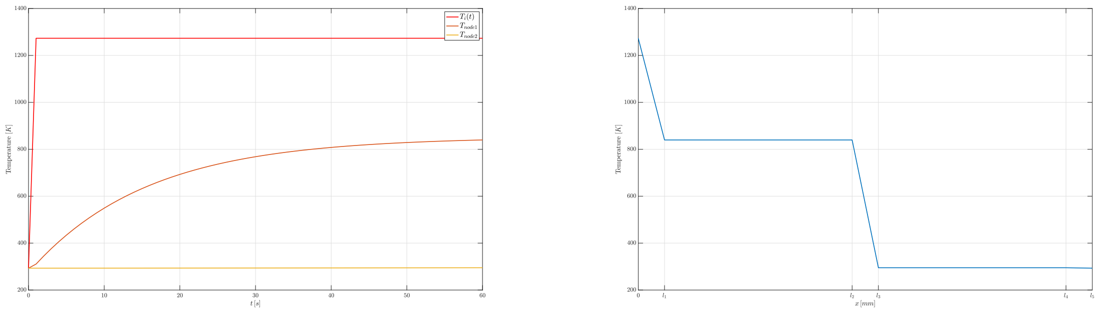
$$\rho l A L \frac{\mathrm{d}T}{\mathrm{d}t} = Q_{in} - Q_{out} \tag{22}$$

Then, recalling that the temperatures at boundaries are know at each instant and that we have only series of conduction resistances, obtaining the governing equations for the 2 nodes case is straightforward:

$$\begin{cases} C_1 \dfrac{\mathrm{d}T_{node1}}{\mathrm{d}t} = \left(R_1 + \dfrac{R_2}{2}\right)^{-1} (T_i - T_{node1}) - \left(\dfrac{R_2}{2} + R_3 + \dfrac{R_4}{2}\right)^{-1} (T_{node1} - T_{node2}) \\ C_2 \dfrac{\mathrm{d}T_{node2}}{\mathrm{d}t} = \left(\dfrac{R_2}{2} + R_3 + \dfrac{R_4}{2}\right)^{-1} (T_{node1} - T_{mode2}) - \left(\dfrac{R_4}{2} + R_5\right)^{-1} (T_{node2} - T_0) \end{cases} \tag{23}$$

The set of equations for the 4 lumped temparatures scenario is extremely similar, and has been omitted here for simplicity and brevity. Numerical integration doesn't provide any complexity, so an explicit method is suitable for the task.
Figure 23 shows the numerical results obtained for the first scenario, while Figure 24 depicts what happens when a total of four lumpings are taken.

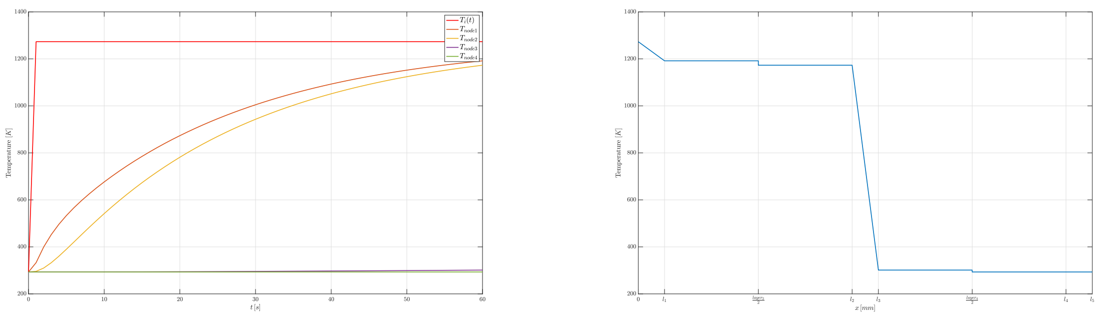
In both cases the behaviour is similar: the conductive layer increases its temperature much more than the insulating one. However, it is also clear that for this specific problem going from two to four total nodes doesn't provide a noticeable step in tha accuracy. A much higher number of lumpings would be required to better approximate the thermal behaviour of the system.



(a) Temperature history

(b) Temperature profile at $t = 60s$

Figure 23: Results for the two node model



(a) Temperature history

(b) Temperature profile at $t = 60s$

Figure 24: Results for the four node model